

## Full Length Article

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## ABSTRACT

Electroencephalography (EEG) captures brain activity and has been widely used in clinic and brain-computer interfaces (BCIs). Classic EEG decoding methods rely on supervised learning, limiting their performance and generalizability. Inspired by the revolutionary impact of large models in other fields, researchers are now investigating EEG foundation models. Recently, state space models (SSMs), such as Mamba, have demonstrated strong sequence modeling capabilities, which may be suitable to model the spatiotemporal dependencies of EEG signals. However, the application of Mamba for EEG representation learning remains largely unexplored. In this paper, we investigate the potential of Mamba for learning generic EEG representations and propose a novel EEG foundation model, EEGMamba. Specifically, we employ Mamba encoder as the backbone of EEGMamba to model the spatiotemporal dependencies among EEG patches. Meanwhile, we use patch-based masked EEG reconstruction to learn generic EEG representations. EEGMamba is pre-trained on a large and diverse EEG corpus (16,724 h) from five datasets. We evaluate EEGMamba on up to six downstream BCI tasks using six public datasets. EEGMamba achieves the state-of-the-art performance across all the tasks, demonstrating its strong capability and generalizability.

## 1. Introduction

Brain-Computer Interfaces (BCIs) (Wu, Pan, & Zheng, 2013; Zhang, Yuan, et al., 2019) enable direct communication between the brain and external devices or computers. Electroencephalography (EEG) is a technique employed to measure and record electrical activity in the brain using electrodes placed on the scalp. EEG is essential for BCIs, providing a non-invasive, real-time measure of brain activity to decode and interpret users' intentions or commands. By analyzing EEG signal patterns and features, algorithms and models have been developed to decode specific brain states (Qi, Liu, Wang, & Pan, 2019), including emotion recognition (Dadebayev, Goh, & Tan, 2022; Wang, Ning, et al., 2024), motor imagery classification (Altaheri et al., 2021; Wang, Yang, et al., 2024), seizure detection (Ahmad et al., 2022; Li, Li et al., 2024), and sleep staging (Einizade, Nasiri, Sardouie, & Clifford, 2023; Phan & Mikkelsen, 2022).

Early EEG decoding studies primarily used traditional machine learning methods (Bashashati, Fatourehchi, Ward, & Birch, 2007; Lotte, Congedo, Lécuyer, Lamarche, & Arnaldi, 2007; McFarland et al., 2006).

With the rapid advancement of deep learning (LeCun, Bengio, & Hinton, 2015), various deep neural networks have been developed to decode EEG signals and address downstream BCI tasks (Craik, He, & Contreras-Vidal, 2019; Pan et al., 2018; Wang et al., 2024b, 2023; Zhou et al., 2025; Zhou, Zhao, Wang, Jiang, Luo, et al., 2024; Zhou, Zhao, Wang, Jiang, Yu, et al., 2024). However, most deep learning models (Lawhern, Solon, Waytowich, Gordon, Hung, & Lance, 2018; Song, Zheng, Liu, & Gao, 2022) rely on supervised learning methods designed for specific tasks or datasets, limiting their generalizability. Improving the generalizability and performance of models remains a significant challenge due to limited data availability and substantial variations in electroencephalography (EEG) signal formats (Yang, Westover, & Sun, 2023). The collection of EEG data is typically task-specific and relatively scarce, as the processes of acquisition and annotation are both costly and time-consuming. Moreover, EEG signal formats exhibit considerable variability across datasets, including differences in channel configurations, temporal resolutions, and recording durations,

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which further complicates the development of robust and generalizable models.

Inspired by the revolutionary impact of large language and vision model, several studies (Foumani, Mackellar, Ghane, Irtza, Nguyen, & Salehi, 2024; Jiang, Zhao, & Lu, 2024; Kostas, Aroca-Ouellette, & Rudzicz, 2021; Yang et al., 2023) have started exploring Transformer-based methods to develop EEG foundation models. These studies utilize Transformers to learn generic EEG representations through self-supervised learning on extensive EEG datasets. The process involves segmenting the original EEG signals into patches across channels, flattening these patches, obtaining patch embeddings, and feeding them into Transformers. This approach models the dependencies among EEG patches, similar to how Vision Transformers handle image patches (Dosovitskiy et al., 2020). While Transformer-based EEG foundation models have achieved good results on downstream datasets, they still require a substantial amount of downstream data for fine-tuning. This reliance on large downstream datasets can be a limitation, especially considering the often limited and difficult-to-collect nature of EEG data. Besides, the quadratic complexity with respect to sequence length in Transformer may become a bottleneck when processing long EEG sequences.

Modern state space models (SSMs) (Gu, Goel, Gupta, & Ré, 2022; Gu, Goel, & Ré, 2021; Gu, Johnson, et al., 2021; Gupta, Gu, & Berant, 2022), derived from the classical Kalman filter, have demonstrated strong sequence modeling capabilities across various tasks and modalities. Mamba (Dao & Gu, 2024; Gu & Dao, 2023), a recent development in SSMs, enhances these capabilities by introducing time-varying parameters and a hardware-aware algorithm for efficient training and inference. While Transformers rely on attention mechanisms to capture long-range dependencies, which can be computationally expensive due to their quadratic complexity with respect to sequence length, Mamba leverages selective state spaces to achieve linear-time sequence modeling. This makes Mamba particularly efficient for processing long sequences, such as EEG signals. EEG data is characterized by complex spatiotemporal patterns that require models capable of understanding intricate dependencies across different channel and time intervals. Mamba's architecture may be effective to model these dependencies, making it a promising alternative to Transformers in EEG foundation models. However, its application to EEG representation learning remains largely unexplored. In this paper, we investigate the potential of selective state space models for learning generic EEG representations and explore a promising road to developing EEG foundation models. Our detailed contributions are as follows:

- We explore the potential of leveraging Mamba to learn generic EEG representations and propose a **novel EEG foundation model**, called **EEGMamba**, for EEG decoding on various clinical and BCI application. To the best of our knowledge, **this is the first study to explore Mamba-based generic EEG representation learning**.
- In order to comprehensively model the spatiotemporal dependencies among EEG patches, we propose a **bidirectional Mamba encoder**, composed of **multiple stacked forward and backward Mamba blocks**, as the backbone of EEGMamba. The forward and backward Mamba blocks are utilized to capture the forward and backward dependencies, respectively.
- EEGMamba is pre-trained on a large EEG corpus (16,724 h) comprising five datasets with diverse formats through patch-based masked EEG reconstruction. We evaluate EEGMamba on up to **six downstream tasks** using **six public datasets**. EEGMamba achieves the state-of-the-art performance across all the tasks, highlighting its **successful capability and generalizability**.

## 2. Related work

### 2.1. EEG decoding

Early studies on EEG decoding predominantly employed traditional machine learning methods (Bashashati et al., 2007; Lotte et al.,

2007; McFarland et al., 2006), which usually rely on hand-crafted features, requiring prior knowledge and lacking generalizability. Deep learning techniques have shifted the focus to methods like Convolution Neural Network (CNN) (Bekhouche, Kajo, Ruichek, & Dornaika, 2022; Ding, Robinson, Zhang, Zeng, & Guan, 2022; Lawhern et al., 2018; Wang, Wang, Qi, Kong, & Wang, 2024), Long Short-Term Memory (LSTM) (Wang, Jiang, Liu, Shang, & Zhang, 2018; Zhang, Shi, et al., 2023), CNN-LSTM (Zhang, Zong, Dou, & Zhao, 2019), Transformer (Song, Jia, Yang, & Xie, 2021), CNN-Transformer (Li, Li et al., 2024; Song et al., 2022; Wang et al., 2024a; Wang, Zhao, Jiang, et al., 2025, 2024), Graph Neural Networks (GNN) (Ding, Robinson, Tong, Zeng, & Guan, 2023; Fu et al., 2025) and etc. These methods perform well on specific tasks or datasets. However, collecting labeled EEG data is costly and labor-intensive, and variations in signal formats across datasets hinder model performance and generalizability.

### 2.2. Brain foundation model

A foundation model (Bommasani et al., 2021) is a deep learning model trained on extensive data using large-scale self-supervision and can be adapted for various downstream tasks through fine-tuning. Recently, there has been a surge in research on Brain Foundation Models. For example, some studies learn EEG representations by utilizing self-supervised learning on CNN (Banville, Chehab, Hyvärinen, Engemann, & Gramfort, 2021). BENDER (Kostas et al., 2021) uses contrastive self-supervised learning on the Transformer backbone to derive generic EEG representations. BrainBERT (Wang et al., 2022) and Brant (Zhang, Yuan, et al., 2023) are Transformer-based foundation model for intracranial signals. Brant-2 (Yuan, Zhang, Chen, Gu, & Yang, 2024) is an enhanced version of Brant, specifically adapted to integrate scalp EEG with intracranial EEG. Brant-X (Zhang, Yuan, Chen, Chen, & Yang, 2024) extends this framework to align EEG with other physiological signals. BIOT (Yang et al., 2023) facilitates joint pre-training and knowledge transfer across diverse biosignal datasets on Linear Transformer. EEG2Rep (Foumani et al., 2024) uses self-prediction for self-supervised EEG representation learning on Transformer. LaBraM (Jiang et al., 2024) learns generic EEG representations by predicting neural tokens of masked EEG patches based on BEiT (Bao, Dong, Piao, & Wei, 2021). EEGPT (Wang, Liu, et al., 2024) is a pretrained transformer model developed for universal EEG feature extraction. CBraMod (Wang, Zhao, Luo, et al., 2025) is an EEG foundation model based on criss-cross transformer. NeuralLM (Jiang, Wang, Lu, & Li, 2025) is an EEG foundation model that leverages the capabilities of Large Language Models (LLMs) by regarding EEG signals as a foreign language. LPSPM (Deng et al., 2024) is a unified flexible large psg model for sleep staging and mental disorder diagnosis based on transformer. Most of the studies use Transformer as the backbone of their EEG foundation models.

### 2.3. State space model

The state space model (SSM) is a traditional method but is limited by its high computational and memory demands. This situation has changed with the advent of Structured State-Space Sequence (S4) model (Gu, Goel, & Ré, 2021), which is a novel alternative to CNNs or Transformers. Fu et al. (2022), Mehta, Gupta, Cutkosky, and Neyshabur (2022), Smith, Warrington, and Linderman (2022) improve the S4 to fills the performance gap between SSMs and Transformer attention. Recently, Gu and Dao (2023) proposes a data-dependent SSM layer and builds a generic language model backbone, Mamba, which is competitive with Transformers at various sizes on large-scale real data and enjoys linear scaling in sequence length. Then a series of Mamba-based models (Li, Singh, & Grover, 2024; Liu et al., 2024; Ma, Li & Wang, 2024; Zhu et al., 2024) were proposed for visual representation learning and various visual applications. Mamba2 (Dao & Gu, 2024) are proposed as a new architecture based on state space duality framework, which is a promising alternative to Transformers. Gui, Chen, Su, Luo,

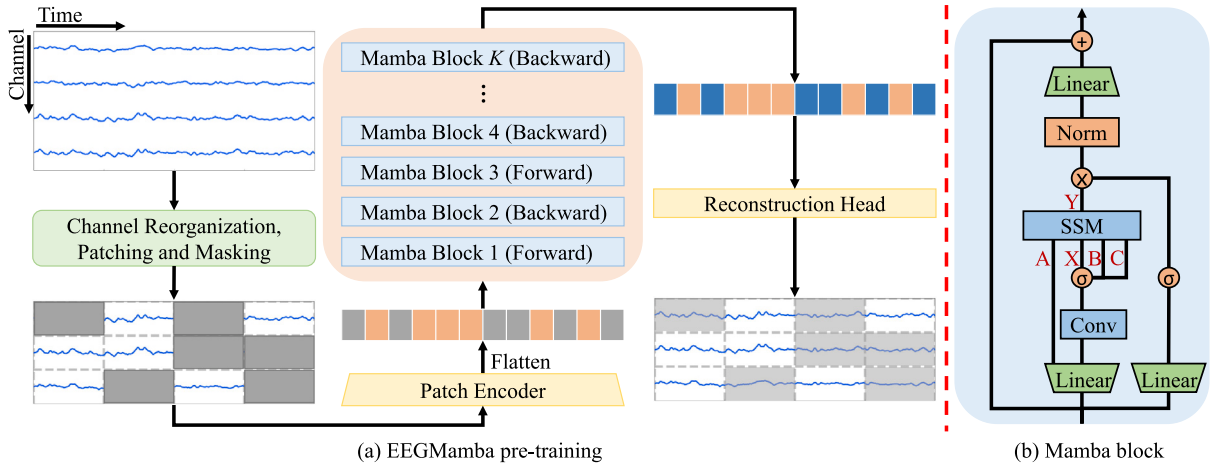


Fig. 1. EEGMamba overview.

and Yang (2024) combines Mamba with mixture of experts for multi-task EEG classification. Bhethanabhotla, Swelam, Siems, Salinas, and Hutter (2024), Ma, Chen, et al. (2024) leverage Mamba to build foundation models for time series forecasting (Chen et al., 2024), proving the potential of Mamba for time series data.

### 3. Method

#### 3.1. Preliminaries

The SSM-based models, i.e., structured state space sequence models (S4) and Mamba, originate from the classic state space model, which maps a 1D input signal  $x(t) \in \mathbb{R}$  to a 1D output signal  $y(t) \in \mathbb{R}$  through a hidden state  $h(t) \in \mathbb{R}^N$ :

$$\begin{aligned} h'(t) &= \mathbf{A}h(t) + \mathbf{B}x(t), \\ y(t) &= \mathbf{C}h(t), \end{aligned} \quad (1)$$

where  $\mathbf{A} \in \mathbb{R}^{N \times N}$  is the evolution parameter,  $\mathbf{B} \in \mathbb{R}^{N \times 1}$  and  $\mathbf{C} \in \mathbb{R}^{1 \times N}$  is the projection parameters.

The S4 and Mamba are the discrete versions of the classic state space model. To deal with the discrete input sequence  $\mathbf{x} = (x_0, x_1, \dots, x_L) \in \mathbb{R}^L$ , they include a timescale parameter  $\Delta$  to transform the continuous parameters  $\mathbf{A}$ ,  $\mathbf{B}$  to discrete parameters  $\bar{\mathbf{A}}$ ,  $\bar{\mathbf{B}}$  by the zero-order hold (ZOH):

$$\begin{aligned} \bar{\mathbf{A}} &= \exp(\Delta \mathbf{A}), \\ \bar{\mathbf{B}} &= (\Delta \mathbf{A})^{-1}(\exp(\Delta \mathbf{A}) - \mathbf{I}) \cdot \Delta \mathbf{B}. \end{aligned} \quad (2)$$

Then the discretized version of Eq. (1) can be reformulated as:

$$\begin{aligned} h_t &= \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t, \\ y_t &= \mathbf{C}h_t. \end{aligned} \quad (3)$$

It is similar to RNN. This recursive process can also be reformulated and computed as a convolution:

$$\begin{aligned} \bar{\mathbf{K}} &= (\mathbf{C}\bar{\mathbf{B}}, \mathbf{C}\bar{\mathbf{A}}\bar{\mathbf{B}}, \dots, \mathbf{C}\bar{\mathbf{A}}^{L-1}\bar{\mathbf{B}}), \\ y &= \mathbf{x} * \bar{\mathbf{K}}, \end{aligned} \quad (4)$$

where  $L$  is the length of the input sequence  $\mathbf{x}$ , and  $\bar{\mathbf{K}} \in \mathbb{R}^L$  is a structured convolutional kernel.

#### 3.2. EEGMamba

The overall architecture of EEGMamba is shown in Fig. 1. We reorganize EEG channels, segment them into patches, and randomly mask some patches. These patches are encoded into embeddings with positional information, flattened into a sequence, and fed into the Mamba encoder to capture spatiotemporal dependencies. Finally, the masked patches are reconstructed from these representations.

##### 3.2.1. Channel reorganization, patching and masking

We denote an original EEG sample as  $S^o \in \mathbb{R}^{C^o \times T}$ , where  $T$  represents the number of timestamps and  $C^o$  represents the number of electrode channels. To enable the model to handle EEG signals with diverse channels, we reorganize the channels of each EEG sample. Specifically, we randomly select  $C$  channels from the  $C^o$  channels of EEG sample  $S^o$  to obtain a new EEG sample  $S \in \mathbb{R}^{C \times T}$ , where  $1 \leq C \leq C^o$ , and shuffle the channel dimension of  $S$ . Then we segment the EEG sample  $S$  with time window length  $t$  to obtain a set of patches  $X \in \mathbb{R}^{C \times n \times t}$ , where  $n = \lfloor \frac{T}{t} \rfloor$  is the number of patches per channel, or the sequence length. Each EEG patch is denoted as  $x \in \mathbb{R}^t$  and the set of patches is  $X = \{x_{i,j} \mid i \in \{1, 2, \dots, C\}, j \in \{1, 2, \dots, n\}\}$ . The total number of patches in  $X$  is  $L = C \cdot n$ . After that, we randomly mask some of them by replacing some patches with a full-zero mask token  $x_M$ . The set of all patches, both masked and unmasked, is denoted as  $\tilde{X} = \{\tilde{x}_{i,j} \mid i \in \{1, 2, \dots, C\}, j \in \{1, 2, \dots, n\}\}$ , where  $\tilde{X} \in \mathbb{R}^{C \times n \times t}$ .

##### 3.2.2. Patch encoder

A patch encoder is used to extract the local features from each patch  $\tilde{x}_{i,j}$  of  $\tilde{X}$ . It consists of three modules: the time-domain, the frequency-domain and the positional module. The time-domain module comprises a one-dimensional convolution layer, a group normalization layer, a GELU activation function and residual connection. We feed each EEG patch  $\tilde{x}_{i,j}$  into this module to obtain a time-domain embedding  $e_{i,j}^t \in \mathbb{R}^d$ , where  $d$  is the embedding dimension. The frequency-domain module utilizes fast Fourier transform (FFT) and a fully-connected layer with residual connection. We apply FFT to extract an energy vector for each patch  $\tilde{x}_{i,j}$  where every dimension indicates the energy of a specific frequency. This vector is then processed through a fully-connected layer to yield the frequency-domain embedding  $e_{i,j}^f \in \mathbb{R}^d$ . We combine the time-domain and frequency-domain embeddings by summing them to obtain  $E = \{e_{i,j} \mid i \in \{1, 2, \dots, C\}, j \in \{1, 2, \dots, n\}\}$ , where  $e_{i,j} \in \mathbb{R}^d$  represents the patch embedding. The positional module incorporates positional information into the patch embeddings. To dynamically encode positional information and adapt to various channel configurations and time lengths, we use conditional positional encoding (CPE) (Chu, Tian, Zhang, Wang, & Shen, 2021). Specifically, we devise a 2D depthwise convolution layer as positional encoder to dynamically generate CPE  $P = \{p_{i,j} \mid i \in \{1, 2, \dots, C\}, j \in \{1, 2, \dots, n\}\}$  from the neighborhoods of each EEG patch, where  $P \in \mathbb{R}^{C \times n \times d}$  and  $p_{i,j} \in \mathbb{R}^d$ . We then add the CPE to the patch embeddings to obtain  $E^p = \{e_{i,j}^p \mid i \in \{1, 2, \dots, C\}, j \in \{1, 2, \dots, n\}\}$ , where  $E^p \in \mathbb{R}^{C \times n \times d}$  is the EEG patch embeddings with positional information.

### 3.2.3. Mamba encoder

We attempt to utilize Mamba for modeling spatiotemporal dependencies among EEG patches. However, like most SSMS, Mamba operates as a unidirectional sequence model, which limits its ability to fully capture these dependencies. To address this limitation, we propose a bidirectional Mamba encoder, composed of multiple stacked forward and backward Mamba blocks, to improve the modeling of spatiotemporal relationships. The forward and backward Mamba blocks are utilized to capture the forward and backward dependencies, respectively. Notably, Mamba has two versions, Mamba1 and Mamba2. We utilize Mamba2 because its larger state space dimension may enhance its ability to model spatiotemporal dependencies. As shown in Fig. 1(b), each Mamba block includes a Mamba2 with a residual connection. We begin by flattening the EEG patch embeddings  $E^p$  along the temporal dimension as follows:

$$E_0 = [e_1, e_2, \dots, e_L], \quad (5)$$

where  $L = C \cdot n$  denotes the number of EEG patches,  $E_0 \in \mathbb{R}^{L \times d}$  is the sequence of EEG patches and  $e_i \in \mathbb{R}^d$  is the  $i$ th EEG patch in  $E_0$ . We then employ stacked forward and backward Mamba blocks to model the sequence of EEG patches, effectively capturing spatiotemporal dependencies. Let  $K$  be the number of Mamba blocks. The bidirectional modeling process is described as follows:

$$E_k = \text{Mamba}_k(E_{k-1}) + E_{k-1}, \quad k \in \{1, 2, \dots, K\}$$

$$\text{Mamba}_k = \begin{cases} \text{Mamba}^f, & k \bmod 2 = 1 \\ \text{Mamba}^b, & k \bmod 2 = 0 \end{cases} \quad (6)$$

where  $\text{Mamba}_k$  is the  $k$ th Mamba block,  $E_k$  is the output of the  $k$ th Mamba block,  $\text{Mamba}^f$  is the forward Mamba block and  $\text{Mamba}^b$  represents the backward Mamba block. The final output of the Mamba encoder is denoted as  $E_K = [e_{1K}, e_{2K}, \dots, e_{LK}]$ , which represents the sequence of EEG patch embeddings, where  $E_K \in \mathbb{R}^{L \times d}$ .

### 3.2.4. Masked EEG reconstruction

Masked autoencoders (MAE) is a simple and effective self-supervised pre-training method for various tasks, including natural language understanding, computer vision, and time series forecasting (Chen et al., 2023; Devlin, Chang, Lee, & Toutanova, 2018; He et al., 2022; Nie, Nguyen, Sinthong, & Kalagnanam, 2022). We utilize patch-based masked EEG reconstruction to learn generic representations from unlabeled EEG signals. Specifically, we use a reconstruction head, consisting of a fully connected layer with a tensor reshaping component, to project the learned EEG representations  $E_K$  into predicted EEG patches  $\hat{X} = \{\hat{x}_{i,j} \mid i \in \{1, 2, \dots, C\}, j \in \{1, 2, \dots, n\}\}$ . Here,  $\hat{x}_{i,j} \in \mathbb{R}^d$  is the predicted EEG patch corresponding to the original patch  $x_{i,j}$ , and  $\hat{X} \in \mathbb{R}^{C \times n \times d}$  is the set of predicted EEG patches corresponding to the original set  $X$ . Consistent with existing MAE studies (He et al., 2022; Xie et al., 2022), we reconstruct only the masked EEG patches. We denote the masked EEG patches as  $X^M$  and their predictions as  $\hat{X}^M$ . The mean squared error (MSE) is used as the reconstruction loss:

$$\mathcal{L}_p = \|\hat{X}^M - X^M\|^2 \quad (7)$$

### 3.3. Task-specific classifier

When applying EEGMamba to downstream EEG decoding tasks, we replace the reconstruction head with a task-specific classifier composed of multilayer perceptrons. The learned EEG representations  $E_K$  is flattened and fed into the classifier to obtain the predicted categories. In the fine-tuning process, the entire model will be fine-tuned and the Cross-Entropy loss function will be used for downstream tasks. The formula of Cross-Entropy loss function is as follows:

$$\mathcal{L} = - \sum_{i=1}^C y_i \cdot \log(\hat{y}_i) \quad (8)$$

where  $C$  is the number of classes,  $y_i$  is the true label for class  $i$  (one-hot encoded), and  $\hat{y}_i$  is the predicted probability for class  $i$ .

## 4. Experiments

### 4.1. Pre-training

#### 4.1.1. Datasets and preprocessing

EEGMamba is pre-trained on an extensive EEG corpus derived from five public datasets with diverse formats, as detailed in Table 1. We selected these datasets due to their abundant data, diverse channel configurations to ensure effective model training, and open-source availability to facilitate further research within the community. These datasets include:

**TUEG (Obeid & Picone, 2016)** comprises a diverse archive of 69,652 clinical EEG recordings from 14,987 subjects across 26,846 sessions, amounting to a total duration of 27,062 h. However, the TUEG dataset is impacted by substantial data contamination, including a considerable amount of unmarked noise, artifacts, and faulty channels. Due to these significant quality issues and high noise levels, we selected clean subsets from the dataset for our study.

**PhysioNet 2018 (Ghassemi et al., 2018; Goldberger et al., 2000)** includes EEG data recorded from 1,985 subjects monitored for sleep disorders. The EEG signals, sampled at 200 Hz, were annotated for six sleep stages and various arousals, with the data split into training (n=994) and test (n=989) sets. Here, we only use the training set.

**Raw EEG Data (Trujillo, 2020)** includes EEG recordings from 64 channels at 256 Hz, collected during the Information-Integration and multidimensional Rule-Based categorization tasks.

**Siena Scalp EEG Database (Detti, Vatti, & Zabalo Manrique de Lara, 2020; Goldberger et al., 2000)** contains EEG recordings from 14 patients, using 31 channels at 512 Hz, with data collected by EB Neuro and Natus Quantum LTM amplifiers and reusable silver/gold cup electrodes.

**B-SNIP1 (Tamminga et al., 2013)** is a specialized dataset designed for studies involving brain-computer interfaces (BCIs) and cognitive neuroscience. It comprises a collection of EEG recordings from subjects engaged in various cognitive tasks, providing a rich source of data for analyzing brain activity patterns associated with different mental processes. We only use the rest-state EEG data. Due to significant quality issues and high noise levels in B-SNIP1, we selected clean subsets from its.

The EEG data is processed with a band-pass filter (0.3 Hz–75 Hz) to remove low- and high-frequency noise and a notch filter (50 or 60 Hz) to eliminate power line interference. All the signals are re-sampled to 200 Hz and segmented into 30-second non-overlapping samples. We normalize the EEG data to 100  $\mu\text{V}$ , ensuring that the values predominantly fall within the range of  $-1$  and  $1$ , consistent with LaBraM. In total, 2,006,883 EEG samples are retained for pre-training, encompassing about 16,724 h, which significantly surpasses the 2,534.78 h used for LaBraM pre-training.

#### 4.1.2. Pre-training settings

EEGMamba was implemented based on the Python 3.10.13 and PyTorch 2.1.1 + CUDA 11.8. The time duration of each EEG patch is 1 s (200 data points). The backbone is a 12-layer Mamba blocks, where the hidden state dimension is 200, the SSM state expansion factor is 64, the head dimension is 50, the local convolution width is 4 and the block expansion factor is 2. We use a 50% mask ratio to randomly mask EEG patches. The batch size is 32, and the number of epochs is 40. The model is trained with the AdamW optimizer using default settings, a learning rate of  $5e-4$ , and a weight decay of  $5e-2$ . CosineAnnealingLR is used to dynamically adjust the learning rate during pre-training. EEGMamba was pre-trained on one machine with Intel Xeon Gold 6226R CPU and one NVIDIA RTX A5000 GPU for about 5 days. The source code and pre-trained weights are publicly available at <https://github.com/wjq-learning/EEGMamba>.



**Table 1**  
Overview of pre-training datasets.

| Datasets                 | Rate   | # Channels | Total duration | Duration | # Samples |
|--------------------------|--------|------------|----------------|----------|-----------|
| TUEG (clean subset)      | 256 Hz | 19         | 9246.2 h       | 30 s     | 1109,545  |
| PhysioNet 2018           | 200 Hz | 6          | 7267.4 h       | 30 s     | 872,089   |
| Raw EEG Data             | 256 Hz | 64         | 42.7 h         | 30 s     | 5125      |
| Siena Scalp EEG Database | 256 Hz | 27         | 140.8 h        | 30 s     | 16,901    |
| B-SNIP1 (clean subset)   | 250 Hz | 60         | 26.9 h         | 30 s     | 3223      |

**Table 2**  
Overview of downstream BCI tasks and datasets.

| BCI Task                                | Datasets     | Rate   | # Channels | Duration | # Samples | Task    |
|-----------------------------------------|--------------|--------|------------|----------|-----------|---------|
| I. Emotion Recognition                  | FACED        | 250 Hz | 32         | 10 s     | 10,332    | 9-class |
| II. Motor Imagery Classification        | PhysioNet-MI | 160 Hz | 64         | 4 s      | 9837      | 4-class |
| III. Sleep Staging                      | ISRUC        | 200 Hz | 6          | 30 s     | 89,240    | 5-class |
| IV. Seizure Detection                   | CHB-MIT      | 256 Hz | 16         | 10 s     | 326,993   | 2-class |
| V. Imagined Speech Classification       | BCIC2020-3   | 256 Hz | 64         | 3 s      | 6000      | 5-class |
| VI. Major Depressive Disorder Diagnosis | MODMA        | 250 Hz | 128        | 15 s     | 1066      | 2-class |

## 4.2. Evaluation

### 4.2.1. Downstream tasks and datasets

To evaluate the performance of EEGMamba, we selected six downstream BCI tasks and their corresponding public datasets:

**I. Emotion Recognition:** FACED (Chen, Wang et al., 2023) is a large, fine-grained affective computing EEG dataset that records 32-channel EEG signals at a 250 Hz sampling rate from 123 subjects. It encompasses nine emotion categories: amusement, inspiration, joy, tenderness, anger, fear, disgust, sadness, and neutral emotion. In our study, subjects 1–80 are used for training, 81–100 for validation, and 101–123 for testing.

**II. Motor Imagery Classification:** PhysioNet-MI (Goldberger et al., 2000; Schalk, McFarland, Hinterberger, Birbaumer, & Wolpaw, 2004) is an EEG motor imagery dataset collected from 109 subjects, utilizing 64 channels at a 160 Hz sampling rate. The dataset includes four motor imagery classes: left fist, right fist, both fists, and both feet. In our study, subject 1–70, 71–89, 90–109 are used for training, validation and test, respectively.

**III. Sleep Staging:** ISRUC (Khalighi, Sousa, Santos, & Nunes, 2016) is a sleep dataset comprising 100 all-night PSG recordings from 100 adults with sleep disorders, some of whom were under the influence of sleep medication. All recordings were scored by sleep experts according to the AASM standard (Iber, Ancoli-Israel, Chesson, & Quan, 2007), with each PSG recording consisting of 6 EEG channels. In our study, subjects 1–80 are used for training, 81–90 for validation, and 91–100 for testing.

**IV. Seizure Detection:** CHB-MIT (Goldberger et al., 2000; Shoeb, 2009) is a database of EEG recordings from 23 pediatric subjects with intractable seizures, monitored after anti-seizure medication withdrawal. The data, recorded using the 10-20 system at 256 Hz, are labeled for seizure detection. We used the standard 16 bipolar montage channels for our analysis. In our study, subjects 1–19 are used for training, 20–21 for validation, and 22–23 for testing.

**V. Imagined Speech Classification:** BCIC2020-3 (Jeong et al., 2022) is a dataset for imagined speech classification, collected from 15 subjects who imagined pronouncing five words/phrases (“hello”, “help me”, “stop”, “thank you”, and “yes”) without vocalizing. EEG signals were recorded at 64 channels with a 256 Hz sampling rate. The dataset provides training, validation, and test sets.

**VI. Major Depressive Disorder Diagnosis:** MODMA (Cai et al., 2022) is a multimodal open dataset designed for mental disorder analysis, primarily including clinically depressed patients alongside matched healthy controls. EEG signals were recorded using 128 electrodes from 53 participants in both resting state and during Dot Probe tasks. In our study, we use only the resting-state EEG data. 15 major depressive disorder (MDD) patients and 18 normal controls (NCs) are used for

training, 4 MDD patients and 5 NCs for validation, and 5 MDD patients and 6 NCs for testing.

**All the datasets used for downstream BCI tasks were collected from different sites or institutions than those used for pre-training.** More details about these tasks and datasets are shown in Table 2. The EEG data is processed with a band-pass filter (0.3 Hz–75 Hz) and a notch filter (50 or 60 Hz). All the signals are resampled to 200 Hz.

### 4.2.2. Fine-tuning settings

We load the pre-trained weights of EEGMamba and replace the reconstruction head with a task-specific classifier composed of multi-layer perceptrons. The learned EEG representations are flattened and fed into this classifier for downstream tasks. For *binary classification*, we use binary cross-entropy (BCE) loss; and for *multi-class classification*, cross-entropy loss. The model is fine-tuned using the AdamW optimizer with default settings, a learning rate of 1e-4 and a weight decay of 5e-2. We also apply CosineAnnealingLR to dynamically adjust the learning rate during fine-tuning. The training is run for 50 epochs with a batch size of 32.

### 4.2.3. Compared methods

To comprehensively evaluate EEGMamba, we compare it with both foundation-model and supervised-learning baselines across all downstream BCI tasks. The foundation baselines are as follows:

**BIOT** (Yang et al., 2023) is an EEG foundation model which learns EEG generic representations through a linear transformer architecture combined with a supervised-unsupervised pre-training strategy.

**LaBraM** (Jiang et al., 2024) is a large brain model that learns generic EEG representations by predicting neural tokens corresponding to masked EEG segments, employing a transformer-based architecture as its backbone. LaBraM is available in three configurations: Base, Large, and Huge. However, only the pre-trained weights for LaBraM-Base are publicly accessible. Consequently, we fine-tune LaBraM-Base exclusively in our experiments.

Following BIOT and LaBraM, the supervised-learning baselines include:

**SPaRCNet** (Jing et al., 2023) is a deep neural network leveraging 1D convolutional neural networks (CNN) with dense residual connections.

**ContraWR** (Yang, Xiao, Westover, & Sun, 2021) is a CNN-based model that first converts biosignals into multi-channel spectrograms, subsequently applying a 2D-CNN-based ResNet (He, Zhang, Ren, & Sun, 2016) to extract features from these spectrograms.

**CNN-Transformer** (Peh, Yao, & Dauwels, 2022) combines a CNN for local feature extraction with a transformer to capture global dependencies.

**FFCL** (Li, Ding, Zhang, & Xiu, 2022) is a neural network that integrates CNN and LSTM in parallel, where the CNN module extracts spatial features and the LSTM module extracts temporal features.

**ST-Transformer** (Song et al., 2021) is a transformer-based model that employs attention mechanisms to learn both spatial and temporal features of EEG signals.

#### 4.2.4. Metrics

For *binary classification*, we use **Balanced Accuracy**, **AUC-PR**, and **AUROC** as evaluation metrics, with AUROC as the monitor score. In a binary classification task, we denote TP as the number of true positive predictions, TN as the number of true negative predictions, FP as the number of false positive predictions and FN as the number of false negative predictions. Balanced accuracy is typically calculated using the following formula:

$$\text{Balanced Accuracy} = \frac{1}{2} \left( \frac{\text{TP}}{\text{TP} + \text{FN}} + \frac{\text{TN}}{\text{TN} + \text{FP}} \right), \quad (9)$$

AUC-PR (Area Under the Precision-Recall Curve) measures the model's performance in detecting the positive class, which is particularly useful for imbalanced datasets. There is no fixed analytical formula for AUC-PR, as it is computed by approximating the area under the Precision-Recall curve. Precision and Recall are typically calculated using the following formulas:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad (10)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad (11)$$

AUC-PR calculation requires computing a series of Precision and Recall points by varying the threshold. These points are used to plot the Precision-Recall curve. Then, AUC-PR can be calculated using numerical integration methods.

AUROC (Area Under the Receiver Operating Characteristic Curve) is the area under the ROC curve, which plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at various threshold settings. TPR and FPR are typically calculated using the following formulas:

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad (12)$$

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}, \quad (13)$$

AUROC calculation requires computing a series of TPR and FPR points by varying the threshold. These points are used to plot the ROC curve. Then, AUROC can be calculated using numerical integration methods.

For *multi-class classification*, we use **Balanced Accuracy**, **Cohen's Kappa**, and **Weighted F1**, with Cohen's Kappa as the monitor score. In a multi-class classification task, we denote  $\text{TP}_i$  as the number of true positives for class  $i$ ,  $\text{FP}_i$  as the number of false positive for class  $i$  and  $\text{FN}_i$  as the number of false negatives for class  $i$ . Balanced accuracy is an average of class-wise recall values, which makes it useful for imbalanced datasets. The formula for Balanced Accuracy in a multi-class setting is as follows:

$$\text{Balanced Accuracy} = \frac{1}{N} \sum_{i=1}^N \frac{\text{TP}_i}{\text{TP}_i + \text{FN}_i}, \quad (14)$$

Cohen's Kappa ( $\kappa$ ) is a statistical measure that evaluates the inter-rater reliability or agreement between predicted and actual labels. Here are the calculation formulas:

$$p_o = \frac{1}{M^2} * \sum_{i=1}^N \sum_{j=1}^N n_{ij}, \quad (15)$$

$$p_e = \frac{1}{M^2} * \sum_{i=1}^N \sum_{j=1}^N n_i * n_j, \quad (16)$$

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \quad (17)$$

where  $M$  is the number of sleep epochs,  $n_{ij}$  is the number of instances where the predicted label is  $i$  and the actual label is  $j$ ,  $n_i$  is the number of epochs with predicted label  $i$ , and  $n_j$  is the number of epochs with actual label  $j$ .

Weighted F1 score is a metric that combines the F1 score for each class, weighted by the support (the number of true instances) of each class. It provides a balanced measure that accounts for both precision and recall, while also giving more importance to classes with more instances. The F1 score for class  $i$  is calculated using the formulas:

$$\text{Precision}_i = \frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i}, \quad (18)$$

$$\text{Recall}_i = \frac{\text{TP}_i}{\text{TP}_i + \text{FN}_i}, \quad (19)$$

$$\text{F1}_i = \frac{2 * \text{Precision}_i * \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i}, \quad (20)$$

The overall weighted F1 score is the average of the F1 scores for all classes, weighted by the support (the number of true instances) for each class  $i$ :

$$\text{Support}_i = \text{TP}_i + \text{FN}_i, \quad (21)$$

$$\text{Weighted F1} = \frac{1}{N} \sum_{i=1}^N \text{Support}_i * \text{F1}_i, \quad (22)$$

### 4.3. Results and analysis

In this section, we present all experimental results and provide an analysis of these findings. Notably, the results are obtained using five different random seeds, and we report both the mean and standard deviation for each metric.

#### 4.3.1. Comparison with existing methods

We conduct a comprehensive comparison of EEGMamba with existing methods across six downstream datasets, the results of which are presented in Table 3. In this table, "Scratch" refers to the version of EEGMamba that is trained on downstream datasets without pre-training.

The comparison clearly highlights that foundation models, in general, outperform traditional supervised learning approaches. Specifically, EEGMamba demonstrates state-of-the-art performance across all six datasets, and it exhibits a lower standard deviation in performance compared to other foundation models such as BIOT and LaBraM. In addition to its superior performance, the model size of EEGMamba is notably smaller than several existing methods, including both supervised learning and foundation model baselines. This efficiency suggests that EEGMamba can effectively and efficiently learn powerful and stable EEG representations without the need for an increased number of parameters, which is a significant advantage in terms of both computational resource requirements and model scalability. Notably, EEGMamba achieves exceptionally strong performance on the FACED and MODMA datasets. Even when trained from scratch, EEGMamba outperforms both BIOT and LaBraM on these datasets, highlighting its superior capability and generalizability. This finding underscores EEGMamba's effectiveness in capturing long-range dependencies within EEG patches, offering a distinct advantage over other models, including Transformers.

#### 4.3.2. Low-resource comparison with other foundation models

In real-world applications, obtaining labeled data for brain-computer interface (BCI) systems or clinical experiments often demands substantial time and financial resources. This challenge highlights the critical importance of developing and exploring EEG foundational models, which have the potential to reduce reliance on large-scale labeled datasets. To demonstrate the practical value of our work, we conduct a comparison of EEGMamba with existing

**Table 3**  
Performance comparison with existing methods (%).

| Methods     | Size  | FACED, 9-class      |                     |                     | PhysioNet-MI, 4-class |                     |                     |
|-------------|-------|---------------------|---------------------|---------------------|-----------------------|---------------------|---------------------|
|             |       | Bal. Acc.           | Kappa <sup>a</sup>  | Weighted F1         | Bal. Acc.             | Kappa <sup>a</sup>  | Weighted F1         |
| SPaRCNet    | 0.79M | 46.73 ± 1.55        | 39.78 ± 2.89        | 47.29 ± 1.33        | 59.32 ± 1.52          | 45.64 ± 2.34        | 59.37 ± 1.47        |
| ContraWR    | 1.6M  | 48.87 ± 0.78        | 42.31 ± 1.51        | 48.84 ± 0.74        | 58.92 ± 1.33          | 45.27 ± 2.48        | 59.18 ± 1.16        |
| CNN-Trans   | 3.2M  | 46.97 ± 1.32        | 40.17 ± 1.68        | 47.20 ± 1.25        | 60.53 ± 1.18          | 47.25 ± 2.23        | 60.41 ± 1.05        |
| FFCL        | 2.4M  | 46.73 ± 1.58        | 39.87 ± 3.83        | 46.99 ± 1.45        | 57.26 ± 0.92          | 43.23 ± 1.82        | 57.01 ± 0.79        |
| ST-Trans    | 3.5M  | 48.10 ± 0.79        | 41.37 ± 1.33        | 47.95 ± 0.96        | 60.35 ± 0.81          | 47.12 ± 1.99        | 60.53 ± 0.75        |
| BIOT        | 3.2M  | 51.18 ± 1.18        | 44.76 ± 2.54        | 51.36 ± 1.12        | 61.53 ± 1.54          | 48.75 ± 2.72        | 61.58 ± 1.97        |
| LaBraM-Base | 5.8M  | 52.73 ± 1.07        | 46.98 ± 1.88        | 52.88 ± 1.02        | 61.73 ± 1.22          | 49.12 ± 1.92        | 61.77 ± 1.41        |
| Scratch     | 3.3M  | 54.02 ± 1.28        | 48.43 ± 2.17        | 54.19 ± 1.03        | 61.08 ± 1.62          | 48.84 ± 2.56        | 61.21 ± 1.43        |
| EEGMamba    | 3.3M  | <b>56.46</b> ± 0.96 | <b>51.34</b> ± 1.27 | <b>56.73</b> ± 0.76 | <b>63.23</b> ± 0.84   | <b>51.31</b> ± 1.24 | <b>63.42</b> ± 0.98 |
| Methods     | Size  | ISRUC, 5-class      |                     |                     | CHB-MIT, 2-class      |                     |                     |
|             |       | Bal. Acc.           | Kappa <sup>a</sup>  | Weighted F1         | Bal. Acc.             | AUC-PR              | AUROC <sup>a</sup>  |
| SPaRCNet    | 0.79M | 74.87 ± 0.75        | 70.97 ± 1.32        | 76.24 ± 0.92        | 58.76 ± 1.91          | 12.47 ± 1.19        | 81.43 ± 1.48        |
| ContraWR    | 1.6M  | 74.02 ± 1.26        | 71.78 ± 1.56        | 76.10 ± 1.37        | 63.44 ± 0.02          | 22.64 ± 1.74        | 80.97 ± 1.14        |
| CNN-Trans   | 3.2M  | 73.63 ± 0.87        | 71.29 ± 1.21        | 77.19 ± 1.05        | 63.89 ± 0.67          | 24.79 ± 2.27        | 86.62 ± 0.82        |
| FFCL        | 2.4M  | 72.77 ± 1.82        | 70.16 ± 2.91        | 76.14 ± 1.97        | 62.62 ± 1.04          | 20.49 ± 3.46        | 82.71 ± 0.51        |
| ST-Trans    | 3.5M  | 73.81 ± 2.05        | 70.13 ± 3.52        | 76.81 ± 1.75        | 59.15 ± 1.95          | 14.22 ± 0.94        | 82.37 ± 4.91        |
| BIOT        | 3.2M  | 75.27 ± 1.21        | 71.92 ± 2.31        | 77.90 ± 1.46        | 70.68 ± 4.57          | 32.77 ± 4.60        | 87.61 ± 2.84        |
| LaBraM-Base | 5.8M  | 76.33 ± 1.02        | 72.31 ± 1.82        | 78.10 ± 1.33        | 70.75 ± 3.58          | 32.87 ± 4.02        | 86.79 ± 1.99        |
| Scratch     | 3.3M  | 76.18 ± 1.45        | 72.25 ± 2.65        | 77.89 ± 1.27        | 70.86 ± 4.87          | 32.99 ± 5.12        | 87.14 ± 2.95        |
| EEGMamba    | 3.3M  | <b>78.99</b> ± 0.93 | <b>74.86</b> ± 1.22 | <b>80.54</b> ± 0.76 | <b>74.45</b> ± 2.46   | <b>38.85</b> ± 4.18 | <b>89.38</b> ± 1.61 |
| Methods     | Size  | BCIC2020-3, 5-class |                     |                     | MODMA, 2-class        |                     |                     |
|             |       | Bal. Acc.           | Kappa <sup>a</sup>  | Weighted F1         | Bal. Acc.             | AUC-PR              | AUROC <sup>a</sup>  |
| SPaRCNet    | 0.79M | 44.26 ± 1.56        | 30.33 ± 2.33        | 44.20 ± 1.08        | 56.37 ± 5.15          | 59.68 ± 4.43        | 59.42 ± 4.08        |
| ContraWR    | 1.6M  | 42.57 ± 1.62        | 30.78 ± 2.18        | 44.07 ± 1.82        | 63.42 ± 4.72          | 68.13 ± 3.89        | 62.07 ± 5.89        |
| CNN-Trans   | 3.2M  | 45.33 ± 0.92        | 31.66 ± 1.18        | 45.06 ± 1.27        | 62.90 ± 5.36          | 66.45 ± 5.17        | 63.77 ± 4.73        |
| FFCL        | 2.4M  | 46.78 ± 1.97        | 33.01 ± 3.59        | 46.89 ± 2.05        | 62.76 ± 3.47          | 67.17 ± 3.71        | 65.08 ± 4.33        |
| ST-Trans    | 3.5M  | 41.26 ± 1.22        | 29.41 ± 1.59        | 42.47 ± 1.38        | 56.46 ± 7.04          | 60.37 ± 4.64        | 58.18 ± 8.42        |
| BIOT        | 3.2M  | 49.20 ± 0.86        | 36.50 ± 1.76        | 49.17 ± 0.79        | 64.89 ± 3.86          | 69.94 ± 3.72        | 66.43 ± 4.71        |
| LaBraM-Base | 5.8M  | 50.60 ± 1.55        | 38.00 ± 2.42        | 50.54 ± 2.05        | 66.81 ± 7.29          | 73.43 ± 5.82        | 68.83 ± 6.22        |
| Scratch     | 3.3M  | 50.13 ± 1.86        | 37.29 ± 2.18        | 49.88 ± 1.69        | 68.07 ± 5.48          | 75.68 ± 4.62        | 71.31 ± 5.79        |
| EEGMamba    | 3.3M  | <b>53.05</b> ± 1.09 | <b>41.83</b> ± 1.57 | <b>53.47</b> ± 0.82 | <b>71.25</b> ± 3.74   | <b>80.67</b> ± 3.48 | <b>73.75</b> ± 3.97 |

<sup>a</sup> Represents the monitor score.

**Table 4**  
Performance comparison on low-resource fine-tuning with existing foundation models (%).

| Methods     | Ratio | FACED               | PhysioNet-MI        | ISRUC               | CHB-MIT             | BCIC2020-3          | MODMA               |
|-------------|-------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|             |       | Kappa               | Kappa               | Kappa               | AUROC               | Kappa               | AUROC               |
| BIOT        | 10%   | 12.69 ± 3.45        | 29.96 ± 3.79        | 59.16 ± 2.39        | 81.45 ± 3.63        | 3.87 ± 2.42         | 48.70 ± 11.26       |
| LaBraM-Base | 10%   | 13.27 ± 3.86        | 31.53 ± 3.41        | 59.35 ± 2.45        | 81.34 ± 2.78        | 5.45 ± 2.87         | 44.95 ± 9.49        |
| EEGMamba    | 10%   | <b>19.73</b> ± 2.56 | <b>36.67</b> ± 2.94 | <b>64.62</b> ± 1.94 | <b>86.43</b> ± 2.21 | <b>10.33</b> ± 2.42 | <b>67.05</b> ± 5.95 |
| BIOT        | 30%   | 25.73 ± 3.46        | 34.77 ± 3.71        | 66.12 ± 2.11        | 85.05 ± 2.98        | 12.49 ± 2.98        | 56.42 ± 9.45        |
| LaBraM-Base | 30%   | 26.72 ± 3.71        | 35.98 ± 3.42        | 66.58 ± 1.84        | 84.17 ± 2.71        | 16.02 ± 3.22        | 55.92 ± 8.77        |
| EEGMamba    | 30%   | <b>32.28</b> ± 2.18 | <b>39.50</b> ± 2.89 | <b>71.23</b> ± 1.69 | <b>87.69</b> ± 1.92 | <b>20.33</b> ± 2.15 | <b>69.14</b> ± 6.78 |
| BIOT        | 50%   | 33.78 ± 2.14        | 40.79 ± 3.31        | 70.45 ± 2.33        | 87.02 ± 3.24        | 19.48 ± 2.06        | 63.71 ± 5.45        |
| LaBraM-Base | 50%   | 35.62 ± 1.88        | 42.14 ± 2.45        | 70.83 ± 1.72        | 85.12 ± 2.41        | 21.33 ± 2.56        | 64.89 ± 8.50        |
| EEGMamba    | 50%   | <b>40.38</b> ± 1.86 | <b>45.05</b> ± 2.06 | <b>73.74</b> ± 1.23 | <b>88.36</b> ± 1.64 | <b>25.67</b> ± 2.19 | <b>72.07</b> ± 5.46 |

**Table 5**  
Performance comparison on different model architectures pre-trained on our pre-training datasets (%).

| Architecture             | FACED               | PhysioNet-MI        | ISRUC               | CHB-MIT             | BCIC2020-3          | MODMA               |
|--------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                          | Kappa               | Kappa               | Kappa               | AUROC               | Kappa               | AUROC               |
| BIOT <sup>•</sup>        | 45.43 ± 2.43        | 49.23 ± 2.19        | 72.34 ± 1.96        | 87.45 ± 2.60        | 37.13 ± 1.89        | 67.17 ± 5.43        |
| LaBraM-Base <sup>•</sup> | 47.96 ± 1.73        | 50.03 ± 1.89        | 73.05 ± 2.06        | 87.56 ± 1.79        | 38.86 ± 2.31        | 69.13 ± 5.03        |
| <i>Our model</i>         |                     |                     |                     |                     |                     |                     |
| - backbone=Transformer   | 47.88 ± 2.31        | 49.73 ± 2.83        | 73.41 ± 2.72        | 87.23 ± 3.46        | 39.14 ± 1.77        | 68.76 ± 5.93        |
| - backbone=Uni-Mamba1    | 48.55 ± 1.83        | 49.04 ± 2.93        | 73.31 ± 1.67        | 86.94 ± 2.18        | 41.55 ± 1.95        | 69.54 ± 5.63        |
| - backbone=Uni-Mamba2    | 49.63 ± 1.08        | 49.55 ± 2.42        | 73.28 ± 2.34        | 87.17 ± 2.71        | 40.87 ± 1.83        | 70.05 ± 4.88        |
| - backbone=Bi-Mamba1     | 50.14 ± 1.56        | <b>51.43</b> ± 1.60 | 74.17 ± 1.53        | 87.63 ± 1.93        | 41.38 ± 1.77        | 72.43 ± 3.77        |
| - backbone=Bi-Mamba2     | <b>51.34</b> ± 1.27 | 51.31 ± 1.24        | <b>74.86</b> ± 1.22 | <b>89.38</b> ± 1.61 | <b>41.83</b> ± 1.57 | <b>73.75</b> ± 3.97 |

foundation models across several datasets where the amount of labeled data is constrained. Specifically, we fine-tune EEGMamba, BIOT, and

LaBraM using varying proportions of labeled data—10%, 30%, and 50%—across all the datasets.

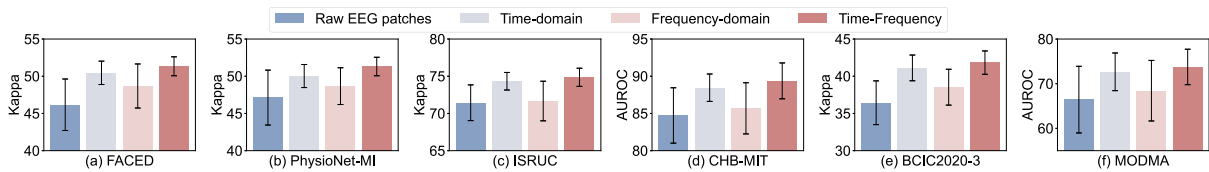


Fig. 2. Ablation study on patch encoder (%).

The results of these experiments are presented in Table 4. It is evident from the results that EEGMamba consistently outperforms existing foundation models in low-resource settings across all datasets. This finding highlights the superior ability of EEGMamba to effectively learn generic EEG representations even when the amount of labeled data is limited. Importantly, when we compare the results presented in Table 4 with those in Table 3, we observe that the performance of EEGMamba in low-resource scenarios is only marginally lower than its performance when trained on the full dataset, particularly on datasets such as PhysioNet-MI, ISRUC, CHB-MIT, and MODMA. This further highlights the practical utility of EEGMamba, showcasing its ability to maintain strong performance even with limited labeled data. Such capabilities provide a significant advantage for real-world applications where labeled resources are often scarce.

#### 4.3.3. Architecture comparison

To further assess the effectiveness of our model architecture, we conducted experiments that compare the performance of existing foundation models with that of our model, using the same pre-training datasets. The experimental setup is outlined as follows:

- (1) BIOT<sup>◆</sup>: we pre-train BIOT from scratch on our pre-training datasets;
- (2) LaBraM-Base<sup>◆</sup>: we pre-train LaBraM-Base from scratch on our pre-training datasets;
- (3) *Our model* with different backbones: we replace the backbone of EEGMamba with different architectures and pre-train each variant on our pre-training datasets.

The results are shown in Table 5. Notably, Bi-Mamba2 achieves the best performance across most downstream datasets, which suggests that the bidirectional Mamba2 architecture serves as an effective backbone for EEG foundation models. While BIOT<sup>◆</sup> and LaBraM<sup>◆</sup> show improved performance compared to their original versions (as reported in Table 3), they still lag significantly behind Bi-Mamba2. This indicates that our model architecture contributes more substantially to performance improvements than simply increasing the size of the pre-training datasets. Bi-Mamba1, while slightly trailing behind Bi-Mamba2, remains highly competitive. This performance difference is likely due to the lower state space dimension in Mamba1 compared to Mamba2, which could limit its expressiveness but still allows it to perform well. In contrast, Uni-Mamba1 and Uni-Mamba2 perform worse than their bidirectional counterparts. This empirical evidence strongly supports our hypothesis that bidirectionality is crucial for effectively modeling the complex spatiotemporal relationships inherent in EEG data. By incorporating both forward and backward Mamba blocks, our architecture can capture more comprehensive contextual information from EEG patches, leading to improved decoding performance. Finally, the Transformer model demonstrates a significant performance gap compared to Mamba-based backbones on the FACHED and MODMA datasets, further highlighting the Transformer's limitations in modeling long-range dependencies. This reinforces the advantage of Mamba-based architectures, which are better equipped to capture the complex, long-range temporal relationships inherent in EEG data.

#### 4.3.4. Ablation study on patch encoder

To evaluate the effectiveness of patch encoding, we conducted an ablation study with the following experimental configurations:

- (1) Raw EEG patches: directly feed raw EEG patches into Mamba Encoder;
- (2) Time-domain: feed raw EEG patches into time-domain module to obtain time-domain features;
- (3) Frequency-domain: feed raw EEG patches into frequency-domain module to obtain frequency-domain features;
- (4) Time-Frequency: combine time-domain and frequency-domain features as used in EEGMamba.

Positional embeddings are added in all configurations to account for the sequential nature of EEG data. The results are shown in Fig. 2.

The Time-Frequency features achieved the best performance across all settings, indicating that the integration of time-domain and frequency-domain features significantly enhances the model's ability to learn powerful and generic EEG representations. This suggests that combining both temporal and frequency information allows the model to capture a more comprehensive view of the EEG signals, ultimately leading to improved performance. In contrast, Raw EEG patches performed the worst, which highlights that the Mamba Encoder alone struggles to effectively model the dependencies present in the original, unprocessed EEG signals. This underscores the importance of time-domain and frequency-domain modules in improving the model's ability. The Time-domain features outperformed the Frequency-domain features, and although they were slightly inferior to the Time-Frequency features, they still demonstrated a notable performance improvement. This result may be due to the inherent presence of frequency-domain information within time-domain signals, as temporal features often contain periodic patterns that reflect underlying frequency characteristics. On the other hand, the Frequency-domain features, derived from the Fast Fourier Transform (FFT) process, are devoid of time-domain information, which may hinder the model's ability to comprehensively capture the information of the EEG signals, leading to relatively lower performance.

#### 4.3.5. Performance comparison of patch overlap in pre-training

In our pre-training process, the patches are non-overlapping. In this section, we investigate the impact of patch overlap by comparing the performance of models trained with and without overlap. Specifically, we configure a 50% overlap between patches in the patch-overlap setting. Under this configuration, we pre-train EEGMamba on the pre-training datasets using the patch-overlap strategy and subsequently fine-tune EEGMamba on the downstream datasets. The results are presented in Table 6. Overall, the findings indicate that there is no significant difference in performance between the non-overlapping and overlapping patch configurations.

#### 4.3.6. Mask ratio analysis

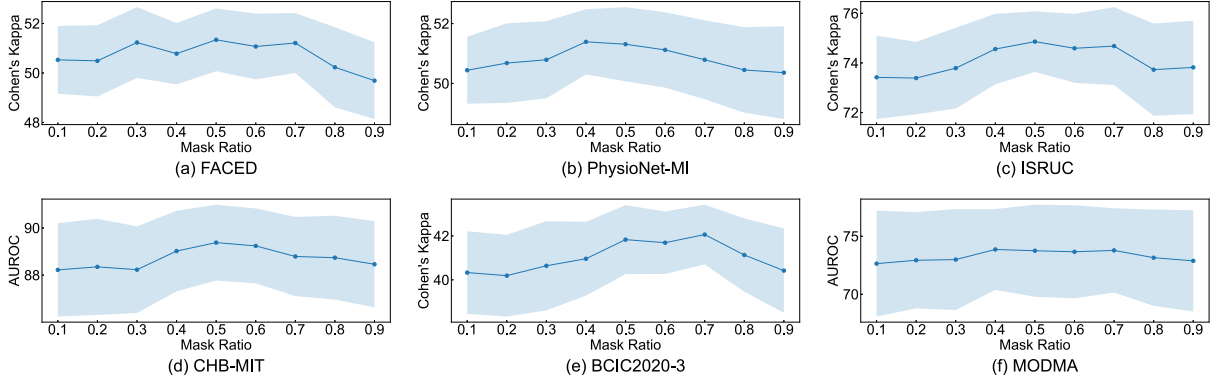
In this section, we conduct a series of comparative experiments to investigate the effect of different mask ratios on EEGMamba's performance across various downstream datasets. The results, summarized in Fig. 3, reveal that EEGMamba generally achieves optimal performance within a mask ratio range of 0.3 to 0.7.

For specific datasets, EEGMamba performs best with a mask ratio of 0.5 on FACHED, ISRUC, and CHB-MIT, as illustrated in Fig. 3(a), (c), and (d), respectively. On PhysioNet-MI, shown in Fig. 3(b), EEGMamba



**Table 6**  
Performance comparison of patch overlap in pre-training (%).

| Architecture  | FACED                   | PhysioNet-MI            | ISRUC                   | CHB-MIT                 | BCIC2020-3              | MODMA                   |
|---------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
|               | Kappa                   | Kappa                   | Kappa                   | AUROC                   | Kappa                   | AUROC                   |
| No overlap    | 51.34 $\pm$ 1.27        | <b>51.31</b> $\pm$ 1.24 | <b>74.86</b> $\pm$ 1.22 | <b>89.28</b> $\pm$ 1.61 | <b>41.83</b> $\pm$ 1.57 | 73.75 $\pm$ 3.97        |
| Patch overlap | <b>51.55</b> $\pm$ 1.36 | 51.26 $\pm$ 1.18        | 74.67 $\pm$ 1.42        | 89.17 $\pm$ 1.49        | 41.75 $\pm$ 1.64        | <b>73.88</b> $\pm$ 4.15 |



**Fig. 3.** Performance comparison on different mask ratio. (%).

**Table 7**  
Parameters, FLOPs and Runtime comparison on CHB-MIT (16 channels, 10 s).

| Methods         | Params | FLOPs  | Runtime |
|-----------------|--------|--------|---------|
| SPaRCNet        | 0.79M  | 65.7M  | 11.5 ms |
| ContraWR        | 1.6M   | 66.4M  | 3.7 ms  |
| CNN-Transformer | 3.2M   | 79.1M  | 16.9 ms |
| FFCL            | 2.4M   | 209.9M | 8.0 ms  |
| ST-Transformer  | 3.5M   | 42.1M  | 3.3 ms  |
| BIOT            | 3.2M   | 483.3M | 6.6 ms  |
| LaBraM-Base     | 5.8M   | 483.0M | 7.8 ms  |
| EEGMamba        | 3.3M   | 186.1M | 6.2 ms  |

achieves its highest performance at a mask ratio of 0.4, with 0.5 following closely behind. For BCIC2020-3, depicted in Fig. 3(e), a mask ratio of 0.7 yields the best results, while 0.5 ranks as the second best. Lastly, on the MODMA dataset, EEGMamba reaches its peak performance with a mask ratio of 0.4, with 0.5 showing comparably strong performance, as shown in Fig. 3(f). These observations suggest that while a mask ratio of 0.5 serves as a universally suitable choice for achieving reliable downstream performance, certain datasets may benefit from fine-tuned mask ratios based on their specific characteristics. However, the consistent strong performance of 0.5 across diverse tasks underscores its practical utility as a default setting, simplifying the application of EEGMamba without requiring extensive parameter tuning for each dataset.

#### 4.3.7. Parameters, FLOPs and runtime comparison

In this section, we perform a detailed comparison of the parameter counts, FLOPs and Runtime between EEGMamba and baseline models, using the CHB-MIT dataset as a representative example. Parameter counts for baseline models are referenced from LaBraM (Jiang et al., 2024), while FLOPs and other values were calculated using Thop.<sup>1</sup> The summarized results, presented in Table 7, reveal that foundation models typically have higher parameter counts and FLOPs than non-foundation baselines. This increase is attributed to the larger model architectures needed to capture generalizable representations. Notably, however, EEGMamba achieves a lower FLOP count compared to similar foundation models, such as BIOT and LaBraM-Base. Meantime,

**Table 8**  
Performance with missing channels on the FACED dataset.

| Methods          | Bal. Acc.               | Kappa                   | Weighted F1             |
|------------------|-------------------------|-------------------------|-------------------------|
| No Missing       | <b>56.46</b> $\pm$ 0.96 | <b>51.34</b> $\pm$ 1.27 | <b>56.73</b> $\pm$ 0.76 |
| Missing Channels | 53.81 $\pm$ 1.05        | 47.79 $\pm$ 1.44        | 54.24 $\pm$ 0.93        |

EEGMamba also achieves a lower Runtime compared to BIOT and LaBraM-Base on the CHB-MIT sample (16 channels, 10 s). This reduction in computational complexity is primarily due to EEGMamba's selective state space modeling strategy, which optimizes efficiency without compromising the model's ability to learn generic EEG representations. Compared to Transformer, Mamba demonstrates superior efficiency when handling long sequences. To evaluate this advantage, we conducted a new runtime comparison on the MODMA dataset (128 channels, 15 s). The runtime for the Transformer-based LaBraM-Base was recorded as 23.0 ms, whereas EEGMamba achieved a runtime of 14.8 ms. This significant reduction in processing time underscores the advantages of EEGMamba in managing extended sequences. Such an EEG foundation model provides a technical basis for real-world EEG systems to efficiently process ultra-long EEG data.

#### 4.3.8. Fine-tuning with missing channels

To provide a more rigorous quantitative analysis of how channel mismatch impacts the fine-tuning process, we conducted a comparative experiment to evaluate the performance of EEGMamba when channels are missing in the FACED dataset. Specifically, we randomly masked half of the channels in the EEG samples from the FACED dataset and utilized the masked data for fine-tuning EEGMamba. The results of this evaluation are presented in Table 8. As shown, the performance with missing channels is approximately 3.0% lower compared to the scenario where no channels are missing. This finding demonstrates that the absence of channels during fine-tuning does lead to a decline in performance; however, the extent of this degradation remains relatively modest.

## 5. Discussion

### 5.1. Implication

This study introduces EEGMamba, an EEG foundation model with implications for advancing generalizable EEG decoding, pioneering

<sup>1</sup> <https://github.com/Lyken17/pytorch-OpCounter>

foundation models in EEG, broadening BCI applications, and guiding future EEG research.

Firstly, EEGMamba addresses the limitations of traditional supervised EEG decoding methods by leveraging Mamba's spatiotemporal modeling capabilities, enabling generalizability across diverse datasets and tasks. Unlike task-specific models, EEGMamba learns versatile representations through pretraining on over 16,000 h of data, setting a benchmark for adaptive EEG models.

Secondly, this work is a critical step toward EEG foundation models. Inspired by large pre-trained models in other fields, EEGMamba's ability to learn generic EEG representations reduces the need for task-specific data, enabling scalable models that can support a variety of EEG-based applications, from cognitive monitoring to neurofeedback.

Moreover, EEGMamba's performance across six BCI tasks demonstrates its potential to serve as a core technology for diverse BCI applications. Traditional BCI systems often lack cross-task adaptability, while EEGMamba's flexibility allows it to handle tasks ranging from emotion recognition to motor imagery, facilitating more adaptive and user-centered BCI systems, especially in areas like neurorehabilitation and medical monitoring.

Finally, EEGMamba provides an effective framework for future EEG signal processing research by employing unsupervised learning strategies. Its patch-based, masked EEG reconstruction method captures EEG signal dynamics without task labels, making it scalable and suitable for large datasets. Future research can build upon EEGMamba's framework to develop more advanced EEG models, guiding the next generation of EEG analysis and BCI applications.

## 5.2. Limitation and future work

One limitation of EEGMamba is its reliance on single-modal EEG data, which may constrain its performance and generalizability. Integrating multimodal data sources, such as EEG combined with other physiological signals, could enhance the model's robustness and applicability, particularly in complex, real-world BCI scenarios. Additionally, although EEGMamba demonstrates strong potential, its practical effectiveness in real-time BCI systems, especially within clinical and neurofeedback contexts, remains unexplored. Besides, while EEGMamba's Mamba-based architecture generally offers efficiency advantages over transformer-based models, these benefits are less pronounced in EEG applications, given the relatively limited number of EEG patches compared to the higher token counts in NLP or high-resolution image patches. Future work will focus on refining the model's architecture to maximize computational efficiency and further reduce resource demands, making EEGMamba more suitable for real-time applications.

## 6. Conclusion

In this paper, we investigate the potential of state space models for learning generic EEG representations and introduce a novel foundation model, EEGMamba. EEGMamba employs Mamba blocks as its backbone to capture the spatiotemporal dependencies of EEG patches using selective state space models. We utilize patch-based masked EEG reconstruction to learn these generic representations. EEGMamba is pre-trained on a large and diverse EEG corpus comprising five datasets. We evaluate EEGMamba on six downstream BCI tasks using six public datasets, employing both full-data and low-resource fine-tuning. EEGMamba consistently achieves state-of-the-art performance across all tasks, proving its strong capability and generalizability.

## CRedit authorship contribution statement

**Jiquan Wang:** Writing – original draft. **Sha Zhao:** Writing – review & editing. **Zhilong Luo:** Validation. **Yangxuan Zhou:** Data curation. **Shijian Li:** Project administration. **Gang Pan:** Project administration.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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